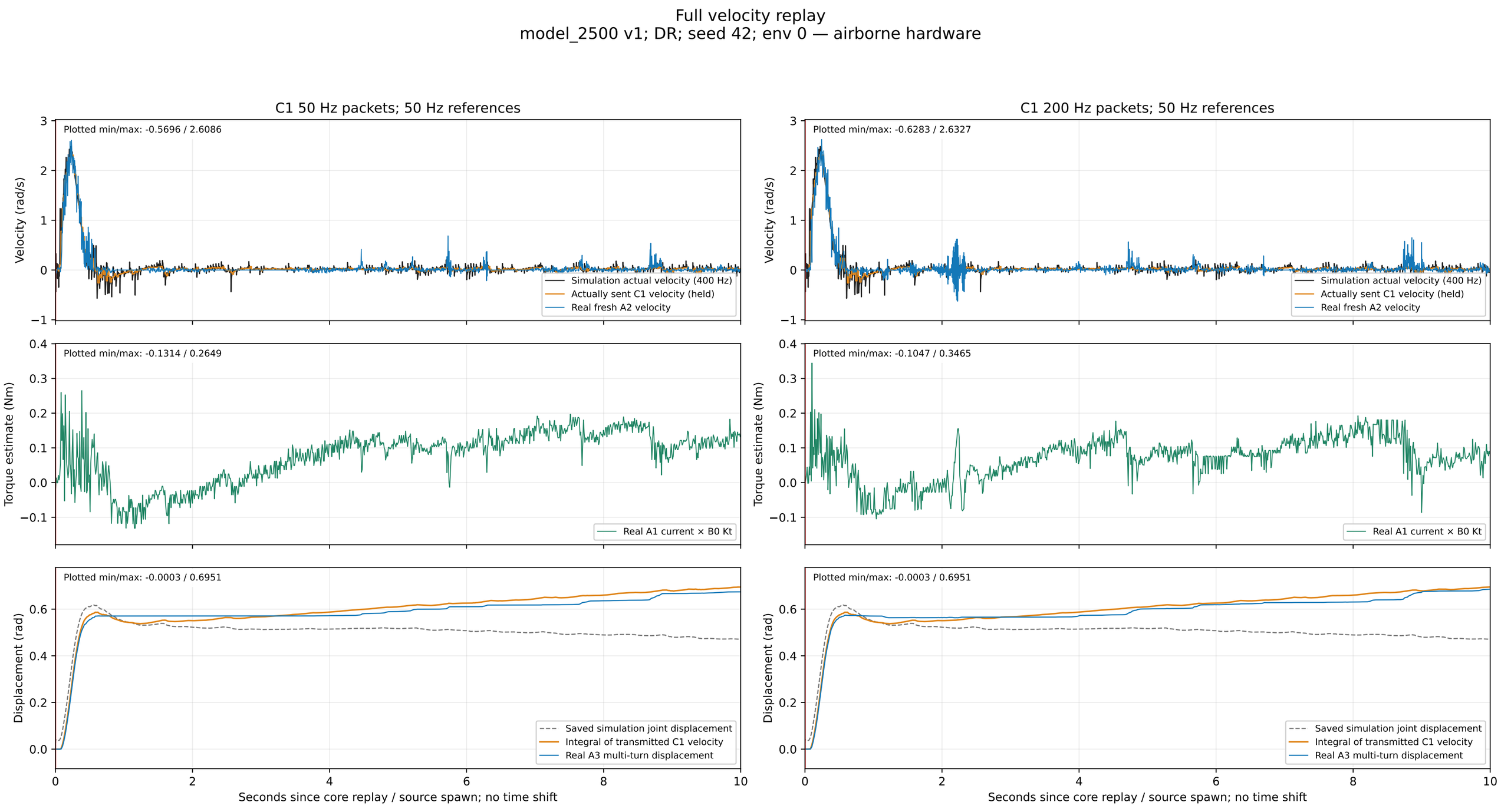
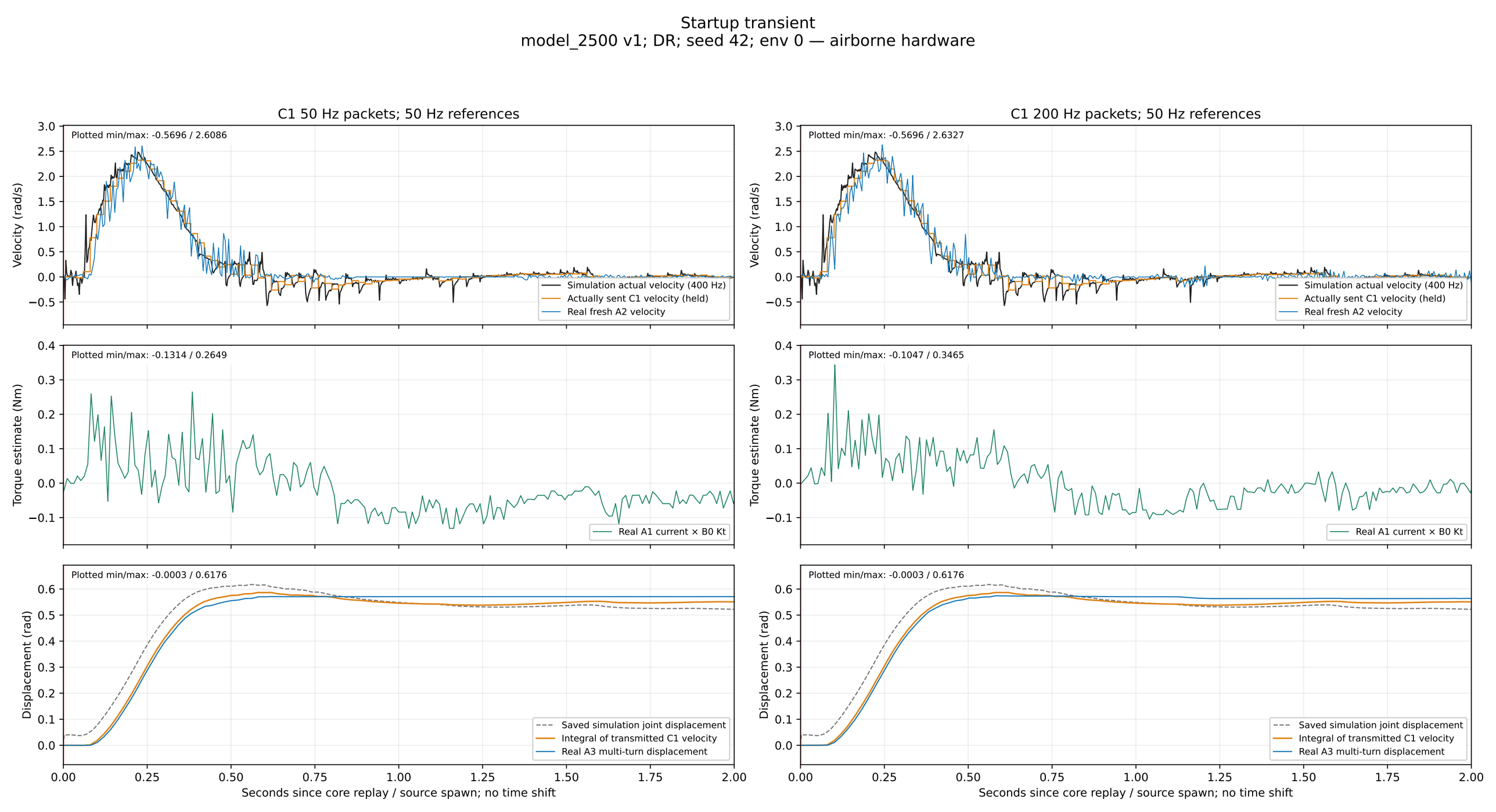
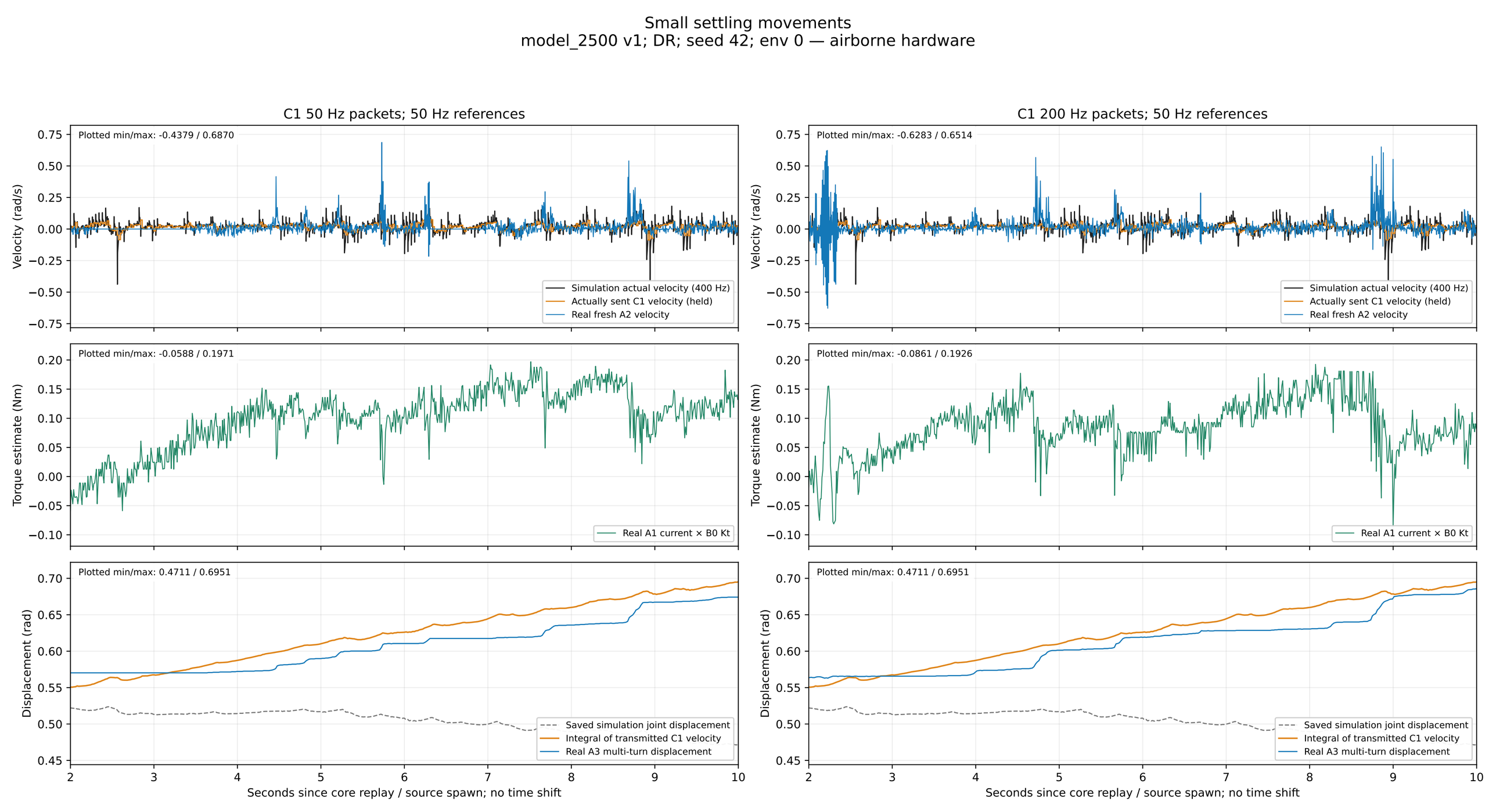
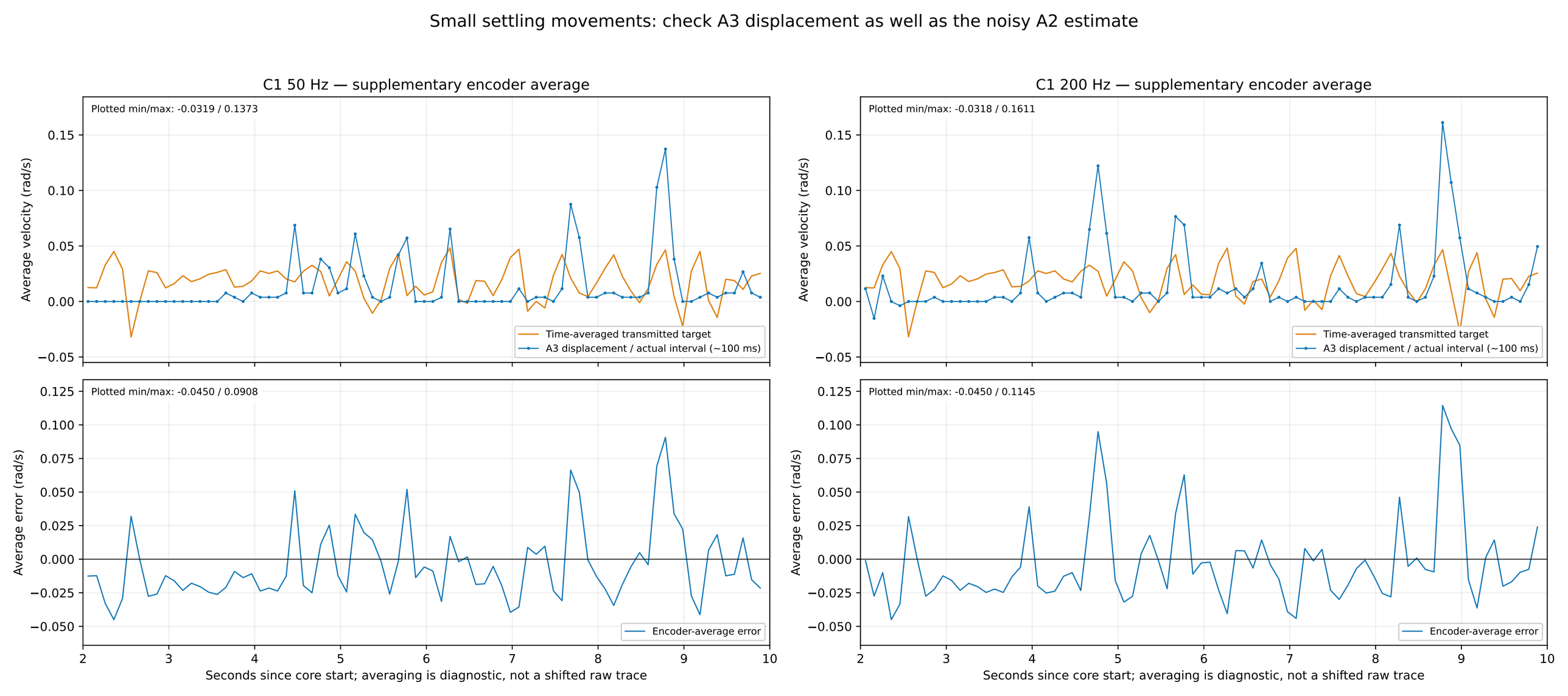
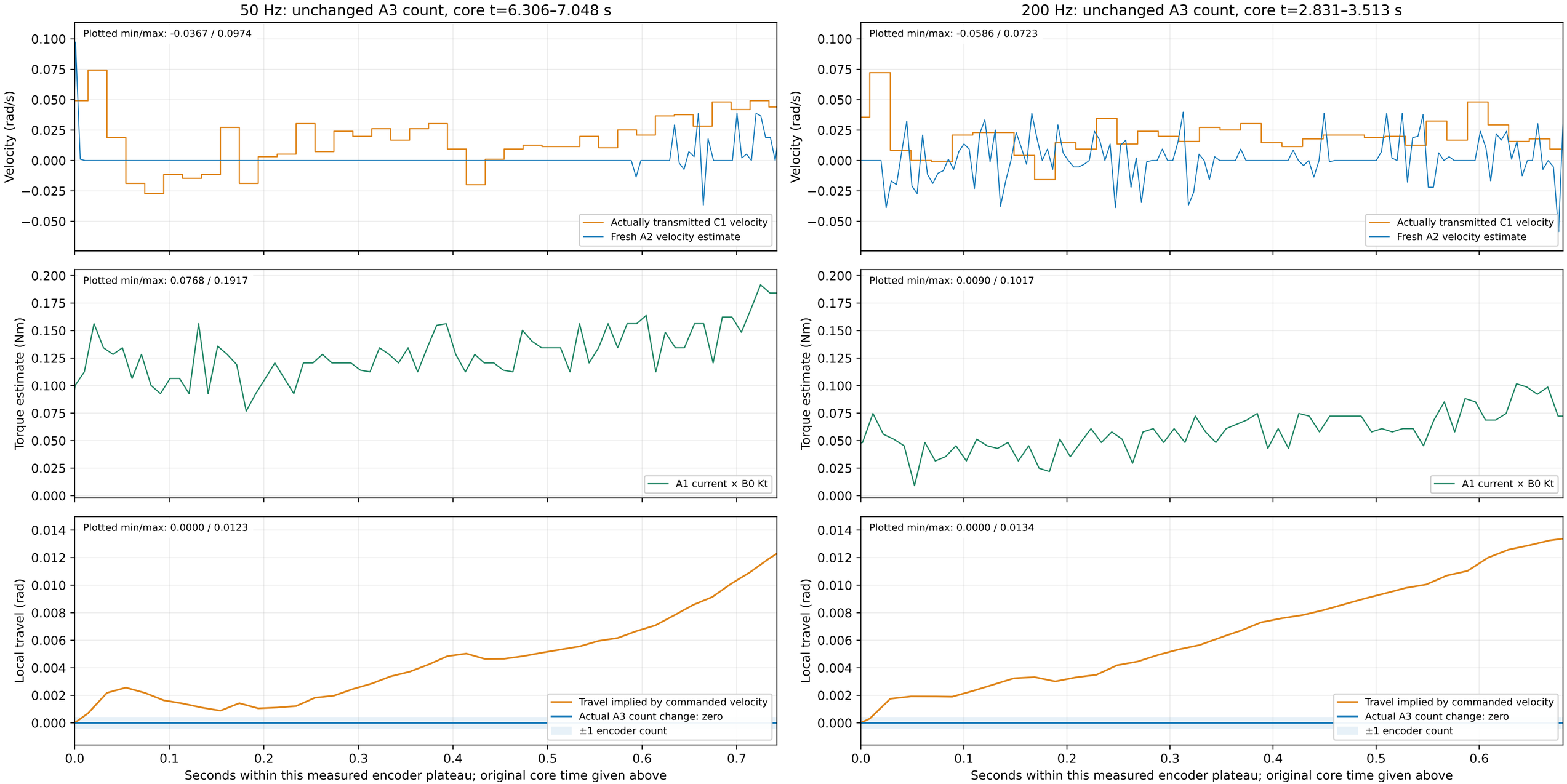
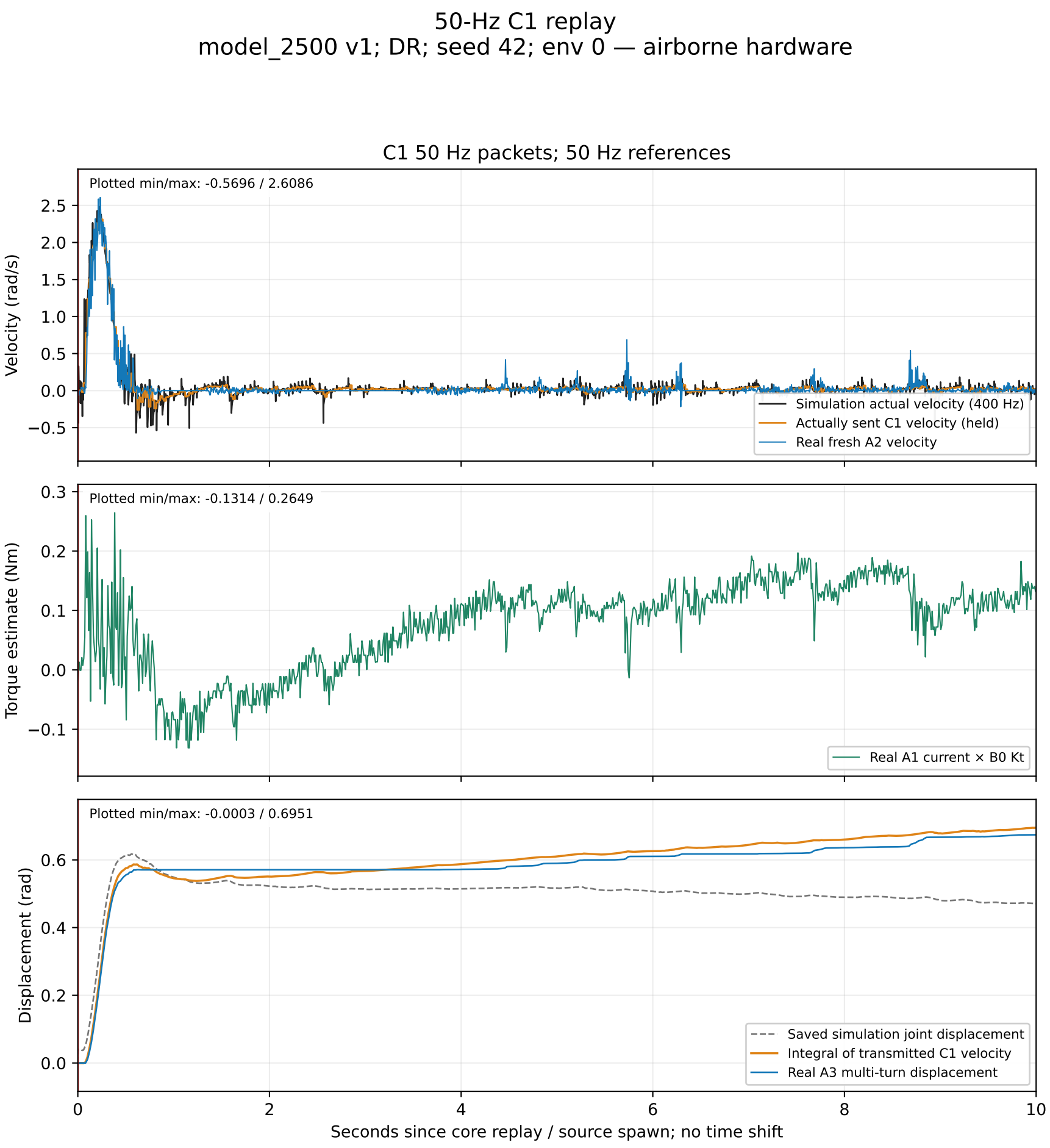

Full velocity replay
model_2500 v1; DR; seed 42; env 0 — airborne hardware
C1 50 Hz packets; 50 Hz references
C1 200 Hz packets; 50 Hz references
Plotted min/max: -0.5696 / 2.6086
Plotted min/max: -0.6283 / 2.6327
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1314 / 0.2649
Plotted min/max: -0.1047 / 0.3465
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0003 / 0.6951
Displacement (rad)
Saved simulation joint displacement
Integral of transmitted C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift

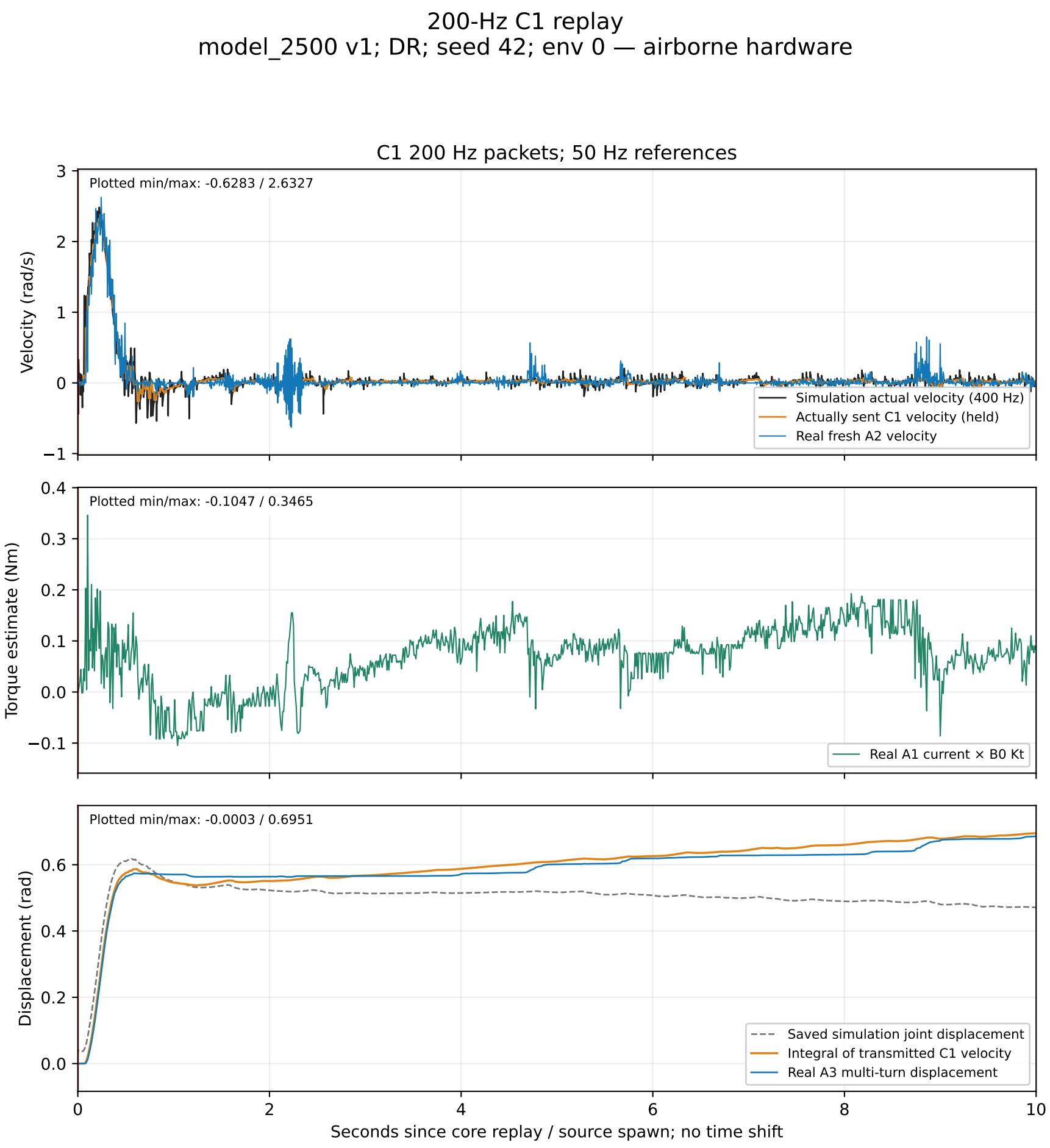

Startup transient
model_2500 v1; DR; seed 42; env 0 — airborne hardware
C1 50 Hz packets; 50 Hz references
C1 200 Hz packets; 50 Hz references
Plotted min/max: -0.5696 / 2.6086
Plotted min/max: -0.5696 / 2.6327
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1314 / 0.2649
Plotted min/max: -0.1047 / 0.3465
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0003 / 0.6176
Displacement (rad)
Saved simulation joint displacement
Integral of transmitted C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift


Small settling movements
model_2500 v1; DR; seed 42; env 0 — airborne hardware
C1 50 Hz packets; 50 Hz references
C1 200 Hz packets; 50 Hz references
Plotted min/max: -0.4379 / 0.6870
Plotted min/max: -0.6283 / 0.6514
Plotted min/max: -0.0588 / 0.1971
Plotted min/max: -0.0861 / 0.1926
Plotted min/max: 0.4711 / 0.6951
Plotted min/max: 0.4711 / 0.6951
Velocity (rad/s)
Torque estimate (Nm)
Displacement (rad)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Real A1 current × B0 Kt
Saved simulation joint displacement
Integral of transmitted C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift
Seconds since core replay / source spawn; no time shift


Small settling movements: check A3 displacement as well as the noisy A2 estimate
C1 50 Hz — supplementary encoder average
Plotted min/max: -0.0319 / 0.1373
Average velocity (rad/s)
Time-averaged transmitted target
A3 displacement / actual interval (~100 ms)
Plotted min/max: -0.0450 / 0.0908
Average error (rad/s)
Encoder-average error
Seconds since core start; averaging is diagnostic, not a shifted raw trace
C1 200 Hz — supplementary encoder average
Plotted min/max: -0.0318 / 0.1611
Average velocity (rad/s)
Time-averaged transmitted target
A3 displacement / actual interval (~100 ms)
Plotted min/max: -0.0450 / 0.1145
Average error (rad/s)
Encoder-average error
Seconds since core start; averaging is diagnostic, not a shifted raw trace


Tiny commands are not followed continuously: position-count plateaus
50 Hz: unchanged A3 count, core t=6.306–7.048 s
200 Hz: unchanged A3 count, core t=2.831–3.513 s
Plotted min/max: -0.0367 / 0.0974
Plotted min/max: -0.0586 / 0.0723
Velocity (rad/s)
Actually transmitted C1 velocity
Fresh A2 velocity estimate
Plotted min/max: 0.0768 / 0.1917
Plotted min/max: 0.0090 / 0.1017
Torque estimate (Nm)
A1 current × B0 Kt
Plotted min/max: 0.0000 / 0.0123
Plotted min/max: 0.0000 / 0.0134
Local travel (rad)
Travel implied by commanded velocity
Actual A3 count change: zero
±1 encoder count
Seconds within this measured encoder plateau; original core time given above
Seconds within this measured encoder plateau; original core time given above


50-Hz C1 replay
model_2500 v1; DR; seed 42; env 0 — airborne hardware
C1 50 Hz packets; 50 Hz references
Plotted min/max: -0.5696 / 2.6086
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1314 / 0.2649
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0003 / 0.6951
Displacement (rad)
Saved simulation joint displacement
Integral of transmitted C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift


200-Hz C1 replay
model_2500 v1; DR; seed 42; env 0 — airborne hardware
C1 200 Hz packets; 50 Hz references
Plotted min/max: -0.6283 / 2.6327
Velocity (rad/s)
Simulation actual velocity (400 Hz)
Actually sent C1 velocity (held)
Real fresh A2 velocity
Plotted min/max: -0.1047 / 0.3465
Torque estimate (Nm)
Real A1 current × B0 Kt
Plotted min/max: -0.0003 / 0.6951
Displacement (rad)
Saved simulation joint displacement
Integral of transmitted C1 velocity
Real A3 multi-turn displacement
Seconds since core replay / source spawn; no time shift